

FEATURE REGISTER

MineOps

Complete Record of All Application Features and Their Status

Group: Syntax Crew

Module: I3691CP

Semester 1, 2026 | Version 1.0 | Last Updated: May 2026

Prepared by: Documentation Team

1. How to Read This Register

This Feature Register is the official record of every feature in the MineOps application. It is maintained by the Documentation Team and updated at each project milestone. Use this document to:

- check what a specific feature does and who it is for.
- find which Firebase service a feature uses.
- see which functional requirement (FR) from the SRS a feature satisfies.
- track whether a feature has been built yet.

Status Key

Status	Meaning
Planned	Feature is defined and approved. Development has not yet started.
In Progress	Feature is currently being coded by the development team.
Completed	Feature is fully built, tested, and working in the running app.
Not Started	Feature is in scope but work has not been planned yet.

2. Feature Register — All 20 Features

This table lists every feature in MineOps with its full description, assigned team, and current build status.

ID	Feature Name	Description	User Roles	Firebase Service	Status	Prior ity	Assigned Team
F-001	User Registration	New users create an account with their name, work email, and password. Firebase Authentication creates the account and assigns a role.	All users	Firebase Auth	Planned	High	Lead Developers
F-002	User Login & Logout	Registered users log in with email and password. Sessions expire after 12 hours of inactivity. Logout clears the local session.	All users	Firebase Auth	Planned	High	Lead Developers
F-003	Role-Based Access Control	Each user sees only the features their role permits. Supervisors can create tasks; Operators cannot. Administrators manage user accounts.	All users	Firebase Auth + Firestore	Planned	High	Firebase Engineers

F-004	Task Creation	Supervisor or Admin fills in: task title, zone, assigned operator(s), priority (Critical/High/Medium/Low), start date, due date, and task category.	Supervisor, Admin	Firestore	Planned	High	Lead Developers
F-005	Task Status Updates	Operators update task status: Not Started → In Progress → Blocked or Completed. Each change is timestamped with the acting user ID.	Operator	Firestore	Planned	High	Lead Developers
F-006	My Tasks List	Operators see a filtered list of only their assigned tasks, ordered by priority and due date. Supervisors can view all tasks across all operators.	Operator, Supervisor	Firestore	Planned	High	Lead Developers
F-007	Raise Hazard Flag	Any user can report a safety hazard by selecting a zone, writing a description, setting severity (Low/Medium/High/Critical), and optionally attaching a photo.	All users	Firestore + Firebase Storage	Planned	High	Lead Developers + Firebase Engineers
F-008	Critical Hazard Alert	When a Critical hazard flag is raised, all supervisors on site receive an immediate push notification. Tasks in the affected zone are automatically put On Hold.	Supervisor	FCM + Firestore	Planned	High	Firebase Engineers
F-009	Hazard Flag List	Shows all active and resolved hazard flags across all zones, filterable by severity and zone. Critical flags appear at the top with a red banner.	All users	Firestore	Planned	High	Lead Developers
F-010	Photo Attachments	Users can attach photos (from camera or gallery) to task updates and hazard reports. Photos are uploaded to Firebase Storage and linked to the relevant record.	All users	Firebase Storage	Planned	Medium	Firebase Engineers
F-011	Live Supervisor Dashboard	Auto-refreshing dashboard (every 5 seconds) showing: task counts by status, active operators and their current tasks, zones with active hazard flags, shift completion %.	Supervisor, Admin	Firestore	Planned	High	Lead Developers
F-012	Overdue Task Escalation	Tasks past their due date that are still open are automatically highlighted red on the dashboard. The supervisor receives a push notification for each overdue task.	Supervisor	Firestore + FCM	Planned	High	Firebase Engineers

F-013	Shift Handover Report	Automatically generated from live Firestore data at shift end. Includes completed tasks, incomplete tasks with reasons, active hazards, equipment notes, and supervisor sign-off.	Supervisor	Firestore	Planned	High	Lead Developers
F-014	Incoming Shift Acknowledgement	Incoming supervisor receives the handover report as a push notification and must tap Acknowledge within 15 minutes. Non-acknowledgement triggers a reminder.	Supervisor	Firestore + FCM	Planned	High	Firebase Engineers
F-015	Push Notifications	Delivers real-time alerts for: new task assignments, task status changes, raised hazard flags, overdue tasks, and shift handover requests.	All users	FCM	Planned	Medium	Firebase Engineers
F-016	Notification Feed	In-app feed showing all notifications received. Unread notifications show a badge count on the tab icon. Tap any notification to navigate to the relevant record.	All users	Firestore	Planned	Medium	Lead Developers
F-017	Compliance Export	Compliance Officer filters tasks, hazard logs, and audit entries by date range, zone, shift, or user, then exports the filtered results as a PDF or CSV file.	Compliance Officer	Firestore	Planned	Medium	Lead Developers
F-018	Audit Log	Immutable record of every system action — task creations, status changes, assignments, hazard flags, and handover sign-offs — with timestamp, acting user, and action type.	Compliance Officer, Admin	Firestore	Planned	High	Firebase Engineers
F-019	Offline Mode	Core features (view tasks, update status, add notes, raise hazard flags) work without internet. Actions are queued locally and synced to Firebase automatically when connectivity returns.	All users	AsyncStorage + Firestore	Planned	Medium	Lead Developers + Firebase Engineers
F-020	Connectivity Status Indicator	A persistent banner at the top of every screen shows: Online (green), Syncing (amber), or Offline (red). The banner also shows how many actions are queued for sync.	All users	AsyncStorage	Planned	Medium	Lead Developers

3. Features by Module

3.1 Authentication Module (F-001 to F-003)

The authentication module is the entry point of the app. It handles everything related to who can get into MineOps and what they are allowed to do once inside. Firebase Authentication manages all login sessions, and Firestore stores the role assigned to each user. The role field on the user record is read immediately after login and controls all subsequent access decisions throughout the app.

3.2 Task Management Module (F-004 to F-006)

Task management is the core module of MineOps. This is where the actual daily work of the mine site is organised. Supervisors create tasks with full details, assign them to operators, and set priority levels. Operators interact with their task list throughout the shift, updating status as they progress. Every status change is immediately written to Firestore so all connected users see the update in real time.

3.3 Hazard & Safety Module (F-007 to F-010)

The hazard module is the safety-critical component of MineOps. It is deliberately designed to be accessible to all user types with minimal steps — any user can raise a hazard flag in three taps. The Critical severity level triggers automated responses: immediate push notifications to all supervisors and automatic task holds in the affected zone. Photo evidence (stored in Firebase Storage) gives supervisors full context when reviewing a flag.

3.4 Dashboard Module (F-011 to F-012)

The supervisor dashboard gives management full real-time visibility of the mine site's operational status. It is the most data-intensive screen in the app, aggregating live reads from the tasks, hazard_flags, and users collections in Firestore. The 5-second auto-refresh interval was chosen as a balance between real-time responsiveness and Firebase read cost.

3.5 Shift Handover Module (F-013 to F-014)

The handover module directly solves the project's stated problem. Instead of a supervisor verbally briefing the incoming shift — or worse, leaving a paper note — MineOps generates a complete structured report from live data. The digital sign-off and acknowledgement flow creates an auditable chain of custody for each shift transition.

3.6 Notifications Module (F-015 to F-016)

Push notifications are delivered via Firebase Cloud Messaging (FCM). FCM is free, reliable, and works even when the MineOps app is closed or running in the background. The in-app notification feed provides a persistent record of all alerts, which is useful for supervisors who receive high volumes of notifications and need to review them at their own pace.

3.7 Compliance Module (F-017 to F-018)

All compliance data in MineOps is stored in an immutable audit log collection. Immutable means records cannot be edited or deleted once written — this is enforced at the Firestore security rules level. The compliance export feature allows Compliance Officers to pull any date range of records, which satisfies OSHA and MSHA digital logging requirements.

3.8 Offline Module (F-019 to F-020)

The offline module uses AsyncStorage (the local device storage library available in React Native) to persist queued actions when the device loses connectivity. When connectivity resumes, a sync routine reads the queue and writes each action to Firestore in order. The connectivity banner is implemented using the [@react-native-community/netinfo](#) library, which listens to the device's network state and updates the banner in real time.